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Caesar Cipher Program – Line by Line Explanation

This document explains the Caesar Cipher Python program line by line and includes a basic security analysis.

Function: caesar_encrypt(text, shift)

This function encrypts the given text by shifting alphabetic characters forward by a fixed number. def caesar_encrypt(text, shift): Defines the encryption function.

- encrypted_text = "": Stores the encrypted output.
- for letter in text: Loops through each character.
- if letter>='a' and letter<='z': Checks for lowercase letters.
- encrypted_text+=chr((ord(letter)+shift-97)%26+97):
    - ord(letter) converts the letter into its ASCII number
    - -97 makes 'a' start from zero
    - + shift moves the letter forward
    - %26 keeps the value within alphabet range
    - +97 converts it back to lowercase
    - chr() converts the number back to a character
    - The encrypted letter is added to encrypted_text
- elif letter>='A' and letter<='Z': Handles uppercase letters.
- encrypted_text+=chr((ord(letter)+shift-65)%26+65): This line encrypts uppercase letters using the same logic as lowercase letters, but:
    - ASCII value 65 is used because 'A' starts at 65
- else: Keeps numbers, spaces, and symbols unchanged.
- return encrypted_text: Returns encrypted message.

Function: caesar_decrypt(ciphertext, shift)

This function reverses the encryption by shifting letters backward.

- def caesar_decrypt(ciphertext, shift): Defines decryption function.
- decrypted_text = "": Stores decrypted output.
- for letter in ciphertext: Loops through encrypted text.
- if letter>='a' and letter <='z': Checks if the character is a lowercase letter.
- decrypted_text += chr((ord(letter)-shift-97)%26+97): This line decrypts lowercase letters:
    - -shift moves the letter backward
    - Other logic is same as encryption
- elif letter>='A' and letter<='Z': Checks if the character is an uppercase letter.
- decrypted_text+=chr((ord(letter)-shift-65)%26+65): Decrypts uppercase letters by shifting them backward.
- else decrypted_text+=letter: Spaces, numbers, and symbols remain unchanged.
- return decrypted_text: Returns original message.

Main Program

- text="I am Mubashir and I have 500$ in my account...": This is the original message that will be encrypted.
- shift=4: This value determines how many positions letters will shift.
- encrypted=caesar_encrypt(text,shift): Calls the encryption function and stores the encrypted

text.

- print("Encrypted Text:",encrypted): Displays the encrypted message.
- decrypted=caesar_decrypt(encrypted,shift): Calls the decryption function to get the original message back.

Security Analysis

Caesar Cipher is a very basic encryption technique. It is not secure for real-world use because it has only 26 possible keys. An attacker can easily try all possible shifts using brute-force attacks. This cipher is mainly used for educational purposes to understand basic encryption concepts.

Code

```python
# Encrypt function
def caesar_encrypt(text,shift):  1 usage
    encrypted_text=""
    for letter in text:
        # For lowercase letters
        if letter>='a' and letter<='z':
            encrypted_text+=chr((ord(letter)+shift-97)%26+97)
        # For uppercase letters
        elif letter>='A' and letter<='Z':
            encrypted_text+=chr((ord(letter)+shift-65)%26+65)
        # For spaces and special characters
        else:
            encrypted_text+=letter
    return encrypted_text
# Decrypt function
def caesar_decrypt(ciphertext,shift):  1 usage
    decrypted_text=""
    for letter in ciphertext:
        if letter>='a' and letter <='z':
            decrypted_text += chr((ord(letter)-shift-97)%26+97)
        elif letter>='A' and letter<='Z':
            decrypted_text+=chr((ord(letter)-shift-65)%26+65)
        else:
            decrypted_text+=letter
    return decrypted_text
```

```python
# Main program
text="I am Mubashir and I have 500$ in my account and My gmail account is malikmubashirhussain482@gmail.com"
shift=4
encrypted=caesar_encrypt(text,shift)
#Encrypt method call
print("Encrypted Text:",encrypted)
#Decrypt method call
decrypted=caesar_decrypt(encrypted,shift)
print("Decrypted Text:",decrypted)
```